

Ctrl + Incision - Sprint 1: Review

Sprint 1 — Advisor Evaluation Summary

Date of Review: December 9, 2025

Advisor: *Dr. Elvis Chen*

Project: *Ctrl + Incision — Image-Guided Surgery*

Team: 5 members

1. Overall Impression

The advisor expressed high confidence in the team's progress and trajectory. Sprint 1 delivered a functional vertical slice demonstrating UI workflow, FakeTracker support, modularized backend components, database logging, and clear team coordination.

The advisor emphasized that Sprint 2 will involve much more technically demanding work, especially the 3D ultrasound (US) reconstruction pipeline, which will require mathematical understanding, careful software design, and substantial development time.

2. Strengths Highlighted by Advisor

✓ Strong Communication & Team Participation

- All members contributed actively during Sprint 1.
- Clear ownership of modules (UI, tracking, DB, refactoring, integration).
- Excellent demo delivery, including background understanding.

✓ Functional & Useful Prototype

- The advisor noted the system is already useful for research and teaching purposes.
- Demonstrated workflow and UI redesign were well received.

✓ Effective Refactoring

- The decision to modularize the CameraController and OverlayRenderer Tracking pipeline was called a “*strong engineering approach.*”

✓ Clear Plan Moving Forward

- Logical breakdown of tasks for Sprint 2 and beyond.
- Good awareness of system architecture and dependencies.

3. Areas for Improvement (Advisor Feedback)

➤ **Hardware Integration Untested (but needed soon)**

- Real tracking hardware (Polaris/Aurora) has not yet been connected.
- Advisor believes it should be “trivial” given current architecture but must be tested early in Sprint 2.

➤ **3D Ultrasound Reconstruction Will Be the Most Difficult Part**

- Advisor highlighted this as the biggest technical challenge.

Will require:

- Real-time performance considerations
- Proper pose–image alignment
- Understanding of transformation chaining
- Error handling & smoothing
- Efficient volume building

➤ **Mathematical Background Needs Strengthening**

- 3D transformation chains
- Hand–eye calibration math
- Probe calibration
- Pose composition + matrix multiplication
- Hough circle detection & RGB → HSV thresholding (tip localization)
- These concepts are crucial before implementing Sprint 2 features.

4. Action Items / Next Steps for Sprint 2

A. More Frequent Meetings

- Advisor recommends weekly check-ins (online or at Robarts).
- Early feedback will prevent mathematical or architectural mistakes.

B. Hardware Integration

- Connect spatial trackers to replace FakeTracker.
- Test ROM files, physical marker tracking, and latency.

C. 3D Reconstruction Pipeline

- Advisor emphasized this must begin early:
- Tracking + ultrasound frame synchronization
- Composing transformation matrices
- Real-time 3D volume insertion

- Optimization for performance
- This task will likely span multiple sprints.

D. Calibration Chain Understanding

- Pivot calibration
- Ultrasound probe calibration
- Hand-eye calibration
- Full transformation graph assembly

E. Explore Vision-Based Tracking Tools

From meeting notes:

- Detect stylus tip using: RGB/Hue thresholding, Hough circle detection
- scikit-image
- Machine learning if needed
- Could support at-home pivot calibration testing.

F. Planning & Documentation

- Maintain a Gantt chart until April 1 (advisor requirement).
- Prepare for lab report + final software submission.

Advisor's Conclusion:

“Overall, I am happy with the current progress. The students are involved and showed good progress. There will be a lot more work needed to be done, especially 3D ultrasound reconstruction, but I have confidence that this team will succeed.”

The team is encouraged to maintain momentum and begin deeper mathematical and algorithmic components in Sprint 2.